

Manually Generating an Si5351 Register Map for 10-MSOP and 20-QFN Devices

1. Introduction

The Si5351 is a highly flexible and configurable clock generator and VCXO. A block diagram of the Si5351 programmable clock IC is shown in Figure 1. To support this flexibility, Skyworks Solutions has created ClockBuilder Pro to create register maps automatically and easily for a given configuration. Since programming with ClockBuilder Pro may not always be well suited for every system's requirements, this document presents the procedures and equations for determining a complete register set from a frequency plan for any 10-MSOP or 20-QFN Si5351 device. Section 2 highlights the overall frequency plan algorithm, and sections 3 and beyond detail all the necessary register calculations.

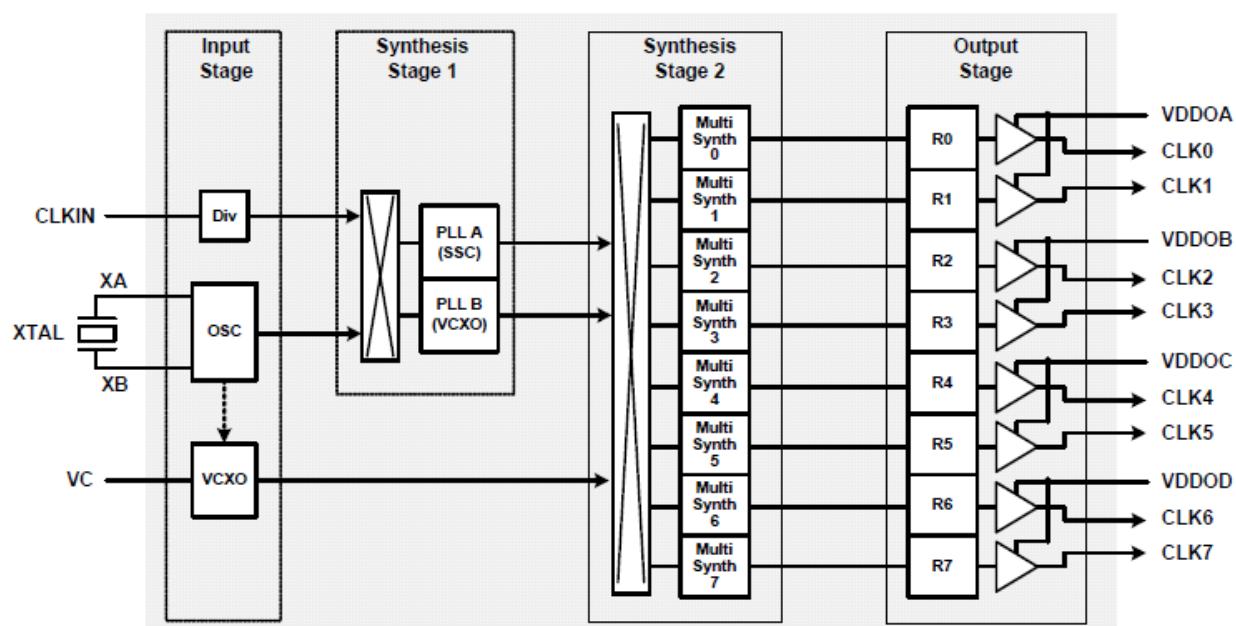


Figure 1. Generalized Si5351 Block Diagram

Figure 1 is a generalized block diagram of the Si5351. The Si5351 is available in three variants:

- Si5351A—XTAL only device.
- Si5351B—XTAL + VCXO.
- Si5351C—XTAL + CLKIN.

If spread spectrum is turned on, it is supported by PLLA.

VCXO functionality (Si5351B devices only) is supported by PLLB.

The Si5351 can also come in three package types: 10-MSOP, 20-QFN, and 16-QFN. This document only covers the programming of 10-MSOP and 20-QFN devices. For more information on programming a 16-QFN device, see AN1234.

2. Conceptualizing a Frequency Plan

The device consists of two PLLs—PLLA and PLLB. Each PLL generates an intermediate VCO frequency in the range of 600 to 900 MHz using a Feedback Multisynth. These VCO frequencies can then be divided down by individual output Multisynth dividers to generate any frequency between 500 kHz and 200 MHz. Additionally, the R dividers can be used to generate any output frequency down to 2.5 kHz. The relationship between the VCO and output frequencies is given below.

$$f_{out_x} = \frac{f_{vco}}{\text{Output_Multisynth}_x \times R_x}$$

$$f_{vco} = f_{in} \times \text{Feedback_Multisynth}$$

Use the steps in the sections below to draw out a conceptual frequency plan map.

2.1. PLL Selection

If spread spectrum is not enabled, either of the two PLLs may be used as the source for any outputs of the Si5351A. If both XTAL and CLKIN input options are used simultaneously (Si5351C only), then one PLL must be reserved for use with CLKIN and one for use with the XTAL.

Note: PLLA must be used for any spread spectrum-enabled outputs. PLLB must be used for any VCXO outputs.

2.1.1. Selecting the Proper VCO Frequencies and Divide Ratios

The general criteria below may be used to set the VCO frequencies. This is a general model, and individual applications may require some modification.

1. Valid Multisynth divider ratios are 4, 6, 8, and any fractional value between $8 + 1/1,048,575$ and 2048. This means that if any output is greater than 112.5 MHz (900 MHz/8), then this output frequency sets one of the VCO frequencies.
2. For the frequencies where jitter is a concern make the output Multisynth divide ratio an integer. If possible, make both output and feedback Multisynth ratios integers.
3. Once criteria 1 and 2 are satisfied, try to select as many integer output Multisynth ratios as possible.

2.2. Output Clock Pin Assignment (Optional)

Some customers may wish to have full control of output clock pin assignment. The guidelines below may be used to assign clocks to pins on the device.

1. Equal output frequencies should share a VDDO bank (e.g., CLK0&1, CLK2&3, etc.) whenever possible in order to ensure minimal jitter due to PCB crosstalk.
2. Isolate unique output frequencies on individual VDDO bank if possible. For example, if the four outputs are 25 MHz, 25 MHz, 27 MHz, and 74.25 MHz, then place them on CLK0, CLK1, CLK2, and CLK4, respectively.
3. Otherwise, use all necessary CLK channels based on the required frequency plan.

Note: The only valid divide ratios for CLK6 and CLK7 Multisynth dividers are even integers between 6 and 254 inclusive. This may limit the two VCO frequencies if all 8 output clocks are used.

3. Configuring Input and PLL Register Parameters (Synthesis Stage 1)

This section describes register parameters related to the input reference and the two PLLs.

3.1. PLL Input Source

The input source for each PLL must be selected. For the Si5351A and Si5351B devices, the only possible source is the XTAL, but for the Si5351C device, each PLL can be synchronized to either an XTAL or a CMOS clock on the CLKIN pin.

3.1.1. XTAL Source

If the source for the PLL is a crystal, PLLx_SRC must be set to 0 in register 15. XTAL_CL[1:0] must also be set to match the crystal load capacitance (see register 183).

3.1.2. CMOS Clock Source

If a PLL needs to be synchronized to a CMOS clock, PLLx_SRC must be 1. The input frequency range of the PLL is 10 to 40 MHz. If CLKIN is > 40 MHz, the CLKIN input divider must be used to bring the PLL input within the 10–40 MHz range. See CLKIN_DIV[1:0], register 15, bits [7:6].

3.2. Feedback Multisynth Divider Equations

Once the input source for each PLL and (if necessary) CLKIN_DIV are determined, the two VCO frequencies selected in Section 2 above can be generated using the following equations. Each feedback divider essentially multiplies the source frequency such that

$$f_{VCO} = f_{XTAL} \times \left(a + \frac{b}{c}\right)$$

and/or

$$f_{VCO} = \frac{f_{CLKIN}}{CLKIN_DIV} \times \left(a + \frac{b}{c}\right)$$

The fractional ratio

$$a + \frac{b}{c}$$

has a valid range of 15 + 0/1,048,575 and 90 and is represented in the Si5351 register space using the equations below.

$$MSNx_P1[17:0] = 128 \times a + \text{Floor}\left(128 \times \frac{b}{c}\right) - 512$$

$$MSNx_P2[19:0] = 128 \times b - c \times \text{Floor}\left(128 \times \frac{b}{c}\right)$$

$$MSNx_P3[19:0] = c$$

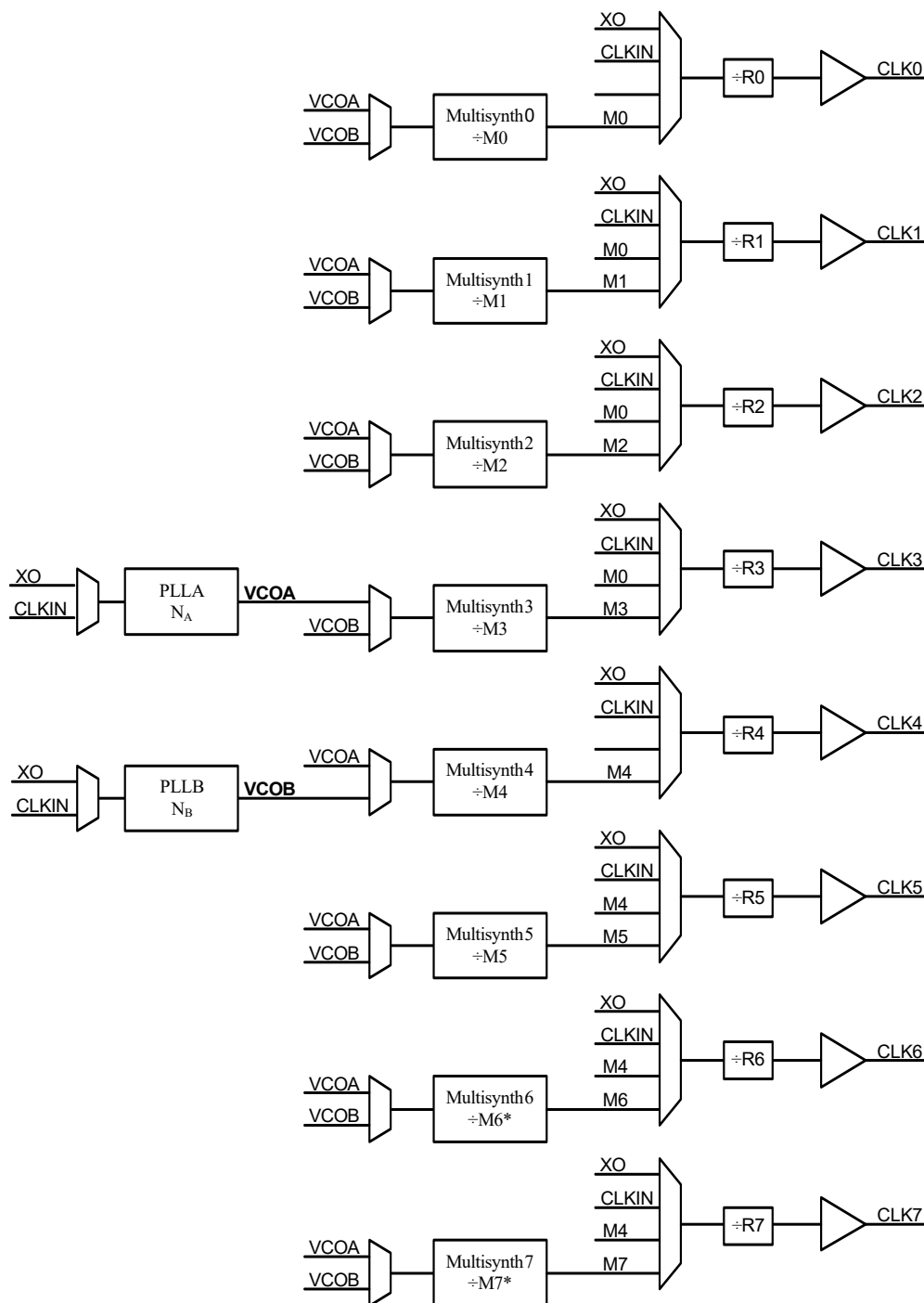
In the equations above, x is used to represent MSNA and MSNB, the PLLA and PLLB Multisynth dividers respectively. The equations above must be repeated for Multisynths MSNA and MSNB. As mentioned earlier in Section “2.1. PLL Selection”, spread spectrum is only supported by PLLA, and the VCXO functionality is only supported by PLLB. When using the VCXO function, set the MSNB divide ratio $a + b/c$ such that $c = 10^6$. This must be taken into consideration when configuring a frequency plan.

3.2.1. Integer Divide Values (FBx_INT)

If $a + b/c$ is an even integer, integer mode may be enabled for PLLA or PLLB by setting parameter FBA_INT or FBB_INT respectively. In most cases setting this bit will improve jitter when using even integer divide values. Whenever spread spectrum is enabled, FBA_INT must be set to 0.

4. Configuring the Output Register Parameters

This section covers the Multisynth and output driver settings necessary for each output. As shown in Figure 2, one of the two VCO frequencies is divided down by each Multisynth divider. This is followed by an output driver state. See “4.2. Output Driver Settings”.



* Multisynth 6 and 7 divide ratios can only be even integers between 6 and 254.

Figure 2. Detailed Dividers and Output Drivers Block Diagram

4.1. Output Multisynth Settings (Synthesis Stage 2)

This section describes the settings related to the individual Multisynths. The Si5351 consists of six fractional Multisynth dividers (MS0-MS5) and two even-integer dividers (MS6 and MS7).

4.1.1. Output Multisynth Source (MSx_SRC)

Each of these dividers can be set to use PLLA or PLLB as its reference by setting MSx_SRC to 0 or 1 respectively. See bit 5 description of registers 16-23.

4.1.2. Output Multisynth Divider Equations ($F_{out} \leq 150 \text{ MHz}$)

Once the PLL source for the output Multisynth is selected, the divide ratio can be set using the equations below. Divider represented as fractional number,

$$a + \frac{b}{c}$$

between $8 + 1/1,048,575$ and 2048.

$$MSx_P1[17:0] = 128 \times a + \text{Floor}\left(128 \times \frac{b}{c}\right) - 512$$

$$MSx_P2[19:0] = 128 \times b - c \times \text{Floor}\left(128 \times \frac{b}{c}\right)$$

$$MSx_P3[19:0] = c$$

In the equations above, $x=0, 1, \dots, 5$. As previously noted, MS6 and MS7 are integer-only dividers. The valid range of values for these dividers is all even integers between 6 and 254 inclusive. For MS6 and MS7, set MSx_P1 directly (e.g., MSx_P1=divide value).

4.1.2.1. Integer Divide Values(MSx_INT)

If any of the MS0-MS5 is an even integer, Multisynth integer mode may be enabled by setting MSx_INT=1 (see registers 16-21, bit 6). In most cases setting this bit will improve jitter when using even integer divide values. Multisynths 6 and 7 inherently operate in integer mode, and so there is no register to turn integer mode on or off. However, it's important to note that Multisynth integer mode cannot be used when adding phase offsets in NVM. In other words, MSx_INT needs to be set to 0 if phase offsets need to be enabled.

4.1.3. Output Multisynth Divider Equations ($150 \text{ MHz} < F_{out} \leq 200 \text{ MHz}$)

Output frequencies greater than 150 MHz are available on Multisynths 0-5. For this frequency range a divide value of 4 must be used by setting

- MSx_P1=0,
- MSx_P2=0,
- MSx_P3=1,
- MSx_INT=1, and
- MSx_DIVBY4[1:0]=11b.

Set the appropriate feedback Multisynth to generate $f_{VCO} = F_{out} \times 4$.

4.2. Output Driver Settings

Once the Multisynth registers are assembled, the output driver settings can be modified according to the information in this section.

4.2.1. Clock Source (CLKx_SRC)

Generally, Multisynth x should be output on CLKx, however XO, CLKIN, or a divided version of either (see section 4.2.2 on R dividers) may also be output on each of the CLKx pins. Additionally, MS0 (or a divided version of MS0) may be output on CLK0-CLK3, and MS4 (or a divided version of MS4) may be output on CLK4-CLK7. See CLKx_SRC description for details.

4.2.2. R Dividers

The R dividers can be used to generate frequencies below about 500 kHz. Each individual output R divider can be set to 1, 2, 4, 8,...128 by writing the proper setting for Rx_DIV. Set this parameter to generate frequencies down to 8kHz.

4.2.3. (CLKx_INV)

In some cases, the user may need to invert the polarity (i.e., 180° phase offset) of one or more outputs with respect to the other outputs. This is achieved by setting CLKx_INV=1.

4.2.4. (CLKx_DIS_STATE)

When a clock is disabled via the OEB pin or OEB control register (reg 3), the output driver may be set to present a logic low, logic high, or high-impedance at the device pin. See CLKx_DIS_STATE description for details.

4.2.5. Unused Clock Outputs

Any unused clock outputs may be powered down to reduce IDD0 current consumption. Set CLKx_PDN=1 to power down unused clock output drivers.

In the case of the 3-output 10-MSOP Si5351, only CLK0, 1, and 2 are used. The remaining clock outputs should be powered down to conserve power. This will be done automatically for any project created in ClockBuilder Pro. However, this will need to be done manually when programming a blank Si5351A-B-GT completely by hand, without the use of ClockBuilder Pro.

5. Configuring Spread Spectrum Register Parameters

Spread spectrum can be enabled on any Multisynth output that uses PLLA as its reference. Valid ranges for spread spectrum include –0.1% to –2.5% down spread and up to $\pm 1.5\%$ center spread. This spread modulation rate is fixed at approximately 31.5 kHz.

The following parameters must be known to properly set up spread spectrum:

- $f_{PFD(A)}$ → input frequency to PLLA in Hz (determined in Sec 2 above and referred to in “3.1.2. CMOS Clock Source”). This is also listed in the ClockBuilder Pro generated register map file as “#PFD(MHz)=...”
- $a + b/c$ → PLLA Multisynth ratio (determined in Sec 2 above).
- $sscAMP$ → Spread amplitude (e.g., for down or center spread amplitude of 1%, $sscAmp = 0.01$).

Use the equations below to set up the desired spread spectrum profile.

Note: Make sure MSNA is set up in fractional mode when using the spread spectrum feature. See parameter FBA_INT in register 22.

5.1. Down Spread

For down spread, four spread spectrum parameters need to be written: SSUDP[11:0], SSDN_P1[11:0], SSDN_P2[14:0], and SSDN_P3[14:0].

Up/Down Parameter:

$$SSUDP[11:0] = \text{Floor}\left(\frac{f_{PFD}}{4 \times 31,500}\right)$$

Intermediate Equation (no register writes):

$$SSDN = 64 \times \left(a + \frac{b}{c}\right) \times \frac{sscAmp}{(1 + sscAmp) \times SSUDP}$$

Down-Spread Parameters:

$$SSDN_P1[11:0] = \text{Floor}[SSDN]$$

$$SSDN_P2[14:0] = 32,767 \times [SSDN - SSDN_P1]$$

$$SSDN_P3[14:0] = 32,767 = 0x7FFF$$

Up-Spread Parameters:

$$SSUP_P1 = 0$$

$$SSUP_P2 = 0$$

$$SSUP_P3 = 1$$

5.2. Center Spread

For center spread, seven spread spectrum parameters need to be written: SSUDP[11:0], SSDN_P1[11:0], SSDN_P2[14:0], SSDN_P3[14:0], SSUP_P1[11:0], SSUP_P2[14:0], and SSUP_P3[14:0].

Up/Down Parameter:

$$\text{SSUDP}[11:0] = \text{Floor}\left(\frac{f_{\text{PFD}}}{4 \times 31,500}\right)$$

Intermediate Equations (no register writes):

$$\text{SSUP} = 128 \times \left(a + \frac{b}{c}\right) \times \frac{\text{sscAmp}}{(1 - \text{sscAmp}) \times \text{SSUDP}}$$

$$\text{SSDN} = 128 \times \left(a + \frac{b}{c}\right) \times \frac{\text{sscAmp}}{(1 + \text{sscAmp}) \times \text{SSUDP}}$$

Up-Spread Parameters:

$$\text{SSUP_P1}[11:0] = \text{Floor}[\text{SSUP}]$$

$$\text{SSUP_P2}[14:0] = 32,767 \times [\text{SSUP} - \text{SSUP_P1}]$$

$$\text{SSUP_P3}[14:0] = 32,767 = 0x7FFF$$

Down-Spread Parameters:

$$\text{SSDN_P1}[11:0] = \text{Floor}[\text{SSDN}]$$

$$\text{SSDN_P2}[14:0] = 32,767 \times [\text{SSDN} - \text{SSDN_P1}]$$

$$\text{SSDN_P3}[14:0] = 32767 = 0x7FFF$$

5.3. Spread Spectrum Enable Pin (SSEN)

The Spread Spectrum Enable control pin is available on the Si5351A and B devices. Spread spectrum enable functionality is a logical OR of the SSEN pin and SSC_EN register bit, so for the SSEN pin to work properly, the SSC_EN register bit must be set to 0.

6. Configuring Initial Phase Offset Register Parameters

Outputs 0-5 of the Si5351 can be programmed with an independent initial phase offset. The phase offset only works when MS0-5 are set as fractional dividers (divider values greater than 8). The phase offset parameter is an unsigned integer where each LSB represents a phase difference of a quarter of the VCO period, $T_{VCO}/4$. Use the equation below to determine the register value. Also, remember that any divider using the phase offset feature needs the MSx_INT bit set to 0.

uint, 1LSB = time $T_{vco}/4$
 T_{vco} is period of VCO/PLL for this output.

$$\text{CLKx_PHOFF}[6:0] = \text{Round}(\text{DesiredOffset}_{(\text{sec})} \times 4 \times F_{VCO})$$

7. Configuring VCXO Parameters (Si5351 B only)

The Si5351B combines free-running clock generation and a VCXO in a single package. The VCXO architecture of the Si5350B eliminates the need for an external pullable crystal. The “pulling” is done at PLLB. Only a standard, low cost, fixed-frequency (25 or 27 MHz) AT-cut crystal is required and is used as the reference source for both PLLA and PLLB.

PLLB must be used as the source for any VCXO output clock. Feedback B Multisynth divider ratio must be set such that the denominator, c, in the fractional divider $a + b/c$ is fixed to 10^6 . Set VCXO_Param register value according to the equation below. Note that 1.03 is a margining factor to ensure the full desired pull range is achieved. For a desired pull-range of ± 30 ppm, the value APR in the equation below is 30, for ± 60 ppm APR is 60, and so on.

$$\text{VCXO_Param}[21:0] = 1.03 \times \left[128a + \frac{b}{10^6} \right] \times \text{APR}$$

8. Si5351 Registers

This section describes the registers and their usage in detail. These values are easily configured using ClockBuilder Pro.

8.1. Register Map Summary

The following is a summary of the register map used to read status, control, and configure the Si5351.

Register	7	6	5	4	3	2	1	0
0	SYS_INIT	LOL_B	LOL_A	LOS_CLKIN	LOS_XTAL	Reserved	REVID[1:0]	
1	SYS_INIT_STKY	LOL_B_STKY	LOL_A_STKY	LOS_CLKIN_STKY	LOS_XTAL_STKY	Reserved		
2	SYS_INIT_MASK	LOL_B_MASK	LOL_A_MASK	LOS_CLKIN_MASK	LOS_XTAL_MASK	Reserved		
3	CLK7_OEB	CLK6_OEB	CLK5_OEB	CLK4_OEB	CLK3_OEB	CLK2_OEB	CLK1_OEB	CLK0_OEB
4–8	Reserved							
9	OEB_MASK7	OEB_MASK6	OEB_MASK5	OEB_MASK4	OEB_MASK3	OEB_MASK2	OEB_MASK1	OEB_MASK0
10–14	Reserved							
15	CLKIN_DIV[1:0]		Reserved		PLLB_SRC	PLLA_SRC	Reserved	
16	CLK0_PDN	MS0_INT	MS0_SRC	CLK0_INV	CLK0_SRC[1:0]		CLK0_IDRV[1:0]	
17	CLK1_PDN	MS1_INT	MS1_SRC	CLK1_INV	CLK1_SRC[1:0]		CLK1_IDRV[1:0]	
18	CLK2_PDN	MS2_INT	MS2_SRC	CLK2_INV	CLK2_SRC[1:0]		CLK2_IDRV[1:0]	
19	CLK3_PDN	MS3_INT	MS3_SRC	CLK3_INV	CLK3_SRC[1:0]		CLK3_IDRV[1:0]	
20	CLK4_PDN	MS4_INT	MS4_SRC	CLK4_INV	CLK4_SRC[1:0]		CLK4_IDRV[1:0]	
21	CLK5_PDN	MS5_INT	MS5_SRC	CLK5_INV	CLK5_SRC[1:0]		CLK5_IDRV[1:0]	
22	CLK6_PDN	FBA_INT	MS6_SRC	CLK6_INV	CLK6_SRC[1:0]		CLK6_IDRV[1:0]	
23	CLK7_PDN	FBB_INT	MS6_SRC	CLK7_INV	CLK7_SRC[1:0]		CLK7_IDRV[1:0]	
24	CLK3_DIS_STATE[1:0]		CLK2_DIS_STATE[1:0]		CLK1_DIS_STATE[1:0]		CLK0_DIS_STATE[1:0]	
25	CLK7_DIS_STATE[1:0]		CLK6_DIS_STATE[1:0]		CLK5_DIS_STATE[1:0]		CLK4_DIS_STATE[1:0]	
26	MSNA_P3[15:8]							
27	MSNA_P3[7:0]							
28	Reserved						MSNA_P1[17:16]	
29	MSNA_P1[15:8]							
30	MSNA_P1[7:0]							
31	MSNA_P3[19:16]				MSNA_P2[19:16]			
32	MSNA_P2[15:8]							
33	MSNA_P2[7:0]							
34	MSNB_P3[15:8]							
35	MSNB_P3[7:0]							
36	Reserved						MSNB_P1[17:16]	
37	MSNB_P1[15:8]							
38	MSNB_P1[7:0]							
39	MSNB_P3[19:16]				MSNB_P2[19:16]			
40	MSNB_P2[15:8]							
41	MSNB_P2[7:0]							
42	MS0_P3[15:8]							
43	MS0_P3[7:0]							
44		R0_DIV[2:0]			MS0_DIVBY4[1:0]		MS0_P1[17:16]	
45	MS0_P1[15:8]							
46	MS0_P1[7:0]							
47	MS0_P3[19:16]				MS0_P2[19:16]			
48	MS0_P2[15:8]							
49	MS0_P2[7:0]							

Register	7	6	5	4	3	2	1	0
50					MS1_P3[15:8]			
51					MS1_P3[7:0]			
52			R1_DIV[2:0]		MS1_DIVBY4[1:0]		MS1_P1[17:16]	
53					MS1_P1[15:8]			
54					MS1_P1[7:0]			
55			MS1_P3[19:16]			MS1_P2[19:16]		
56					MS1_P2[15:8]			
57					MS1_P2[7:0]			
58					MS2_P3[15:8]			
59					MS2_P3[7:0]			
60			R2_DIV[2:0]		MS2_DIVBY4[1:0]		MS2_P1[17:16]	
61					MS2_P1[15:8]			
62					MS2_P1[7:0]			
63			MS2_P3[19:16]			MS2_P2[19:16]		
64					MS2_P2[15:8]			
65					MS2_P2[7:0]			
66					MS3_P3[15:8]			
67					MS3_P3[7:0]			
68			R3_DIV[2:0]		MS3_DIVBY4[1:0]		MS3_P1[17:16]	
69					MS3_P1[15:8]			
70					MS3_P1[7:0]			
71			MS3_P3[19:16]			MS3_P2[19:16]		
72					MS3_P2[15:8]			
73					MS3_P2[7:0]			
74					MS4_P3[15:8]			
75					MS4_P3[7:0]			
76			R4_DIV[2:0]		MS4_DIVBY4[1:0]		MS4_P1[17:16]	
77					MS4_P1[15:8]			
78					MS4_P1[7:0]			
79			MS4_P3[19:16]			MS4_P2[19:16]		
80					MS4_P2[15:8]			
81					MS4_P2[7:0]			
82					MS5_P3[15:8]			
83					MS5_P3[7:0]			
84			R5_DIV[2:0]		MS5_DIVBY4[1:0]		MS5_P1[17:16]	
85					MS5_P1[15:8]			
86					MS5_P1[7:0]			
87			MS5_P3[19:16]			MS5_P2[19:16]		
88					MS5_P2[15:8]			
89					MS5_P2[7:0]			
90					MS6_P1[7:0]			
91					MS7_P1[7:0]			
92	Reserved		R7_DIV[2:0]		Reserved		R6_DIV[2:0]	
149	SSC_EN				SSDN_P2[14:8]			
150					SSDN_P2[7:0]			
151	SSC_MODE				SSDN_P3[14:8]			
152					SSDN_P3[7:0]			
153					SSDN_P1[7:0]			
154			SSUDP[11:8]				SSDN_P1[11:8]	
155					SSUDP[7:0]			
156							SSUP_P2[14:8]	

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Register	7	6	5	4	3	2	1	0
157	SSUP_P2[7:0]							
158		SSUP_P3[14:8]						
159	SSUP_P3[7:0]							
160	SSUP_P1[7:0]							
161	SS_NCLK[3:0]				SSUP_P1[11:8]			
162	VCXO_Param[7:0]							
163	VCXO_Param[15:8]							
164	Reserved		VCXO_Param[21:16]					
165	Reserved	CLK0_PHOFF[6:0]						
166	Reserved	CLK1_PHOFF[6:0]						
167	Reserved	CLK2_PHOFF[6:0]						
168	Reserved	CLK3_PHOFF[6:0]						
169	Reserved	CLK4_PHOFF[6:0]						
170	Reserved	CLK5_PHOFF[6:0]						
171–176	Reserved							
177	PLLB_RST	Reserved	PLLA_RST	Reserved				
178–182	Reserved							
183	XTAL_CL		Reserved					
184–186	Reserved							
187	CLKIN_FA-NOUT_EN	XO_FA-NOUT_EN	Reserved	MS_FA-NOUT_EN	Reserved			
188-255	Reserved							

9. Register Descriptions

Register 0. Device Status

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	SYS_INIT	LOL_B	LOL_A	LOS_CLKIN	LOS_XTAL	Reserved	REVID[1:0]	
Type	R	R	R	R	R	R	R	

Reset value = 0000 0000

Bit	Name	Function
7	SYS_INIT	System Initialization Status. During power up the device copies the content of the NVM into RAM and performs a system initialization. The device is not operational until initialization is complete. It is not recommended to read or write registers in RAM through the I ² C interface until initialization is complete. An interrupt will be triggered (INTR pin = 0, Si5351C only) during the system initialization period. 0: System initialization is complete. Device is ready. 1: Device is in system initialization mode.
6	LOL_B	PLL B Loss Of Lock Status. Si5351A/C only. PLL B will operate in a locked state when it has a valid reference from CLKIN or XTAL. A loss of lock will occur if the frequency of the reference clock forces the PLL to operate outside of its lock range or if the reference clock fails to meet the minimum requirements of a valid input signal as specified in the Si5351 data sheet. An interrupt will be triggered (INTR pin = 0, Si5351C) during a LOL condition. 0: PLL B is locked. 1: PLL B is unlocked. When the device is in this state it will trigger an interrupt causing the INTR pin to go low (Si5351C only).
5	LOL_A	PLL A Loss Of Lock Status. PLL A will operate in a locked state when it has a valid reference from CLKIN or XTAL. A loss of lock will occur if the frequency of the reference clock forces the PLL to operate outside of its lock range as specified in the data sheet, or if the reference clock fails to meet the minimum requirements of a valid input signal as specified in the Si5351 data sheet. An interrupt will be triggered (INTR pin = 0, Si5351C only) during a LOL condition. 0: PLL A is operating normally. 1: PLL A is unlocked. When the device is in this state it will trigger an interrupt causing the INTR pin to go low (Si5351C only).
4	LOS_CLKIN	CLKIN Loss Of Signal (Si5351C Only). A loss of signal status indicates if the reference clock fails to meet the minimum requirements of a valid input signal as specified in the Si5351 data sheet. An interrupt will be triggered (INTR pin = 0, Si5351C only) during a LOS condition. 0: Valid clock signal at the CLKIN pin. 1: Loss of signal detected at the CLKIN pin.

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3	LOS_XTAL	Crystal Loss of Signal A loss of signal status indicates that the crystal failed to meet the minimum requirements of a valid input signal as specified in the Si5351 datasheet. This bit going high will trigger an interrupt (INTR pin = 0) on a Si5351C device. 0: Valid crystal signal at the XA and XB Pins 1: Loss of crystal signal detected
2	Reserved	Leave as default.
1:0	REVID[1:0]	Revision number of the device.

Register 1. Interrupt Status Sticky

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	SYS_INIT_STKY	LOL_B_STKY	LOL_A_STKY	LOS_- CLKIN_STKY	LOS_ XTAL_STKY			
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Reset value = 0000 0000

Bit	Name	Function
7	SYS_INIT_STKY	System Calibration Status Sticky Bit. The SYS_INIT_STKY bit is triggered when the SYS_INIT bit (register 0, bit 7) is triggered high. It remains high until cleared. Writing a 0 to this register bit will cause it to clear. 0: No SYS_INIT interrupt has occurred since it was last cleared. 1: A SYS_INIT interrupt has occurred since it was last cleared.
6	LOL_B_STKY	PLLB Loss Of Lock Status Sticky Bit. The LOL_B_STKY bit is triggered when the LOL_B bit (register 0, bit 6) is triggered high. It remains high until cleared. Writing a 0 to this register bit will cause it to clear. 0: No PLL B interrupt has occurred since it was last cleared. 1: A PLL B interrupt has occurred since it was last cleared.
5	LOL_A_STKY	PLLA Loss Of Lock Status Sticky Bit. The LOL_A_STKY bit is triggered when the LOL_A bit (register 0, bit 5) is triggered high. It remains high until cleared. Writing a 0 to this register bit will cause it to clear. 0: No PLLA interrupt has occurred since it was last cleared. 1: A PLLA interrupt has occurred since it was last cleared.
4	LOS_CLKIN_STKY	CLKIN Loss Of Signal Sticky Bit (Si5351C Only). The LOS_CLKIN_STKY bit is triggered when the LOS_CLKIN bit (register 0, bit 4) is triggered high. It remains high until cleared. Writing a 0 to this register bit will cause it to clear. 0: No LOS_CLKIN interrupt has occurred since it was last cleared. 1: A LOS_CLKIN interrupt has occurred since it was last cleared.
3	LOS_XTAL_STKY	Crystal Loss of Signal Sticky Bit The LOS_XTAL_STKY bit is triggered when the LOS_XTAL bit (register 0, bit 3) is triggered high. It remains high until cleared. Writing a 0 to this register bit will cause it to clear. 0: No LOS_XTAL interrupt has occurred since it was last cleared. 1: A LOS_XTAL interrupt has occurred since it was last cleared.
2:0	Reserved	Leave as default.

Register 2. Interrupt Status Mask

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	SYS_INIT_MASK	LOL_B_MASK	LOL_A_MASK	LOS_- CLKIN_MASK	LOS_ XTAL_MASK			
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Reset value = 0000 0000

Bit	Name	Function
7	SYS_INIT_MASK	System Initialization Status Mask. Use this mask bit to prevent the INTR pin (Si5351C only) from going low when SYS_INIT is asserted. 0: Do not mask the SYS_INIT interrupt. 1: Mask the SYS_INIT interrupt.
6	LOL_B_MASK	PLL B Loss Of Lock Status Mask. Use this mask bit to prevent the INTR pin (Si5351C only) from going low when LOL_B is asserted. 0: Do not mask the LOL_B interrupt. 1: Mask the LOL_B interrupt.
5	LOL_A_MASK	PLL A Loss Of Lock Status Mask. Use this mask bit to prevent the INTR pin (Si5351C only) from going low when LOL_A is asserted. 0: Do not mask the LOL_A interrupt. 1: Mask the LOL_A interrupt.
4	LOS_- CLKIN_MASK	CLKIN Loss Of Signal Mask (Si5351C Only). Use this mask bit to prevent the INTR pin (Si5351C only) from going low when LOS_-CLKIN is asserted. 0: Do not mask the LOS_CLKIN interrupt. 1: Mask the LOS_CLKIN interrupt.
3	LOS_ XTAL_MASK	Crystal Loss of Signal Mask Use this mask bit to prevent the INTR pin (Si5351C only) from going low when LOS_XTAL is asserted. 0: Do not mask the LOS_XTAL interrupt 1: Mask the LOS_XTAL interrupt
2:0	Reserved	Leave as default.

Register 3. Output Enable Control

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	CLK7_OEB	CLK6_OEB	CLK5_OEB	CLK4_OEB	CLK3_OEB	CLK2_OEB	CLK1_OEB	CLK0_OEB
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Reset value = 0000 0000

Bit	Name	Function
7:0	CLKx_OEB	Output Disable for CLKx. Where x = 0, 1, 2, 3, 4, 5, 6, 7 0: Enable CLKx output. 1: Disable CLKx output.

Register 9. OEB Pin Enable Control Mask

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	OEB_MAS_K7	OEB_MAS_K6	OEB_MAS_K5	OEB_MAS_K4	OEB_MAS_K3	OEB_MAS_K2	OEB_MAS_K1	OEB_MAS_K0
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Reset value = 0000 0000

Bit	Name	Function
7:0	OEB_CLKx	OEB pin enable control of CLKx. Where x = 0, 1, 2, 3, 4, 5, 6, 7 0: OEB pin controls enable/disable state of CLKx output. 1: OEB pin does not control enable/disable state of CLKx output.

Register 15. PLL Input Source

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	CLKIN_DIV[1:0]				PLLB_SRC	PLLA_SRC		
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Reset value = 0000 0000

Bit	Name	Function
7:6	CLKIN_DIV[1:0]	CLKIN Input Divider. Valid PLL input range is 10-40 MHz. If CLKIN is > 40 MHz, CLKIN input divider, CLKIN_DIV, must be used to bring the PLL input within the 10-40 MHz range. 00b: Divide by 1. 01b: Divide by 2. 10b: Divide by 4. 11b: Divide by 8.
5:4	Reserved	Leave as default.
3	PLLB_SRC	Input Source Select for PLLB. 0: Select the XTAL input as the reference clock for PLLB. 1: Select the CLKIN input as the reference clock for PLLB (Si5351C only).
2	PLLA_SRC	Input Source Select for PLLA. 0: Select the XTAL input as the reference clock for PLLA. 1: Select the CLKIN input as the reference clock for PLLA (Si5351C only).
1:0	Reserved	Leave as default.

Register 16. CLK0 Control

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	CLK0_PDN	MS0_INT	MS0_SRC	CLK0_INV	CLK0_SRC[1:0]		CLK0_IDRV[1:0]	
Type	R/W	R/W	R/W	R/W	R/W		R/W	

Reset value = 0000 0000

Bit	Name	Function
7	CLK0_PDN	Clock 0 Power Down. This bit allows powering down the CLK0 output driver to conserve power when the output is unused. 0: CLK0 is powered up. 1: CLK0 is powered down.
6	MS0_INT	MultiSynth 0 Integer Mode. This bit can be used to force MS0 into Integer mode to improve jitter performance. Note that the fractional mode is necessary when a delay offset is specified for CLK0. 0: MS0 operates in fractional division mode. 1: MS0 operates in integer mode.
5	MS0_SRC	MultiSynth Source Select for CLK0. 0: Select PLLA as the source for MultiSynth0. 1: Select PLLB (Si5351A/C only) or VCXO (Si5351B only) for MultiSynth0.
4	CLK0_INV	Output Clock 0 Invert. 0: Output Clock 0 is not inverted. 1: Output Clock 0 is inverted.
3:2	CLK0_SRC[1:0]	Output Clock 0 Input Source. These bits determine the input source for CLK0. 00: Select the XTAL as the clock source for CLK0. This option by-passes both synthesis stages (PLL/VCXO & MultiSynth) and connects CLK0 directly to the oscillator which generates an output frequency determined by the XTAL frequency. 01: Select CLKIN as the clock source for CLK0. This by-passes both synthesis stages (PLL/VCXO & MultiSynth) and connects CLK0 directly to the CLKIN input. This essentially creates a buffered output of the CLKIN input. 10: Reserved. Do not select this option. 11: Select MultiSynth 0 as the source for CLK0. Select this option when using the Si5351 to generate free-running or synchronous clocks.
1:0	CLK0_IDRV[1:0]	CLK0 Output Rise and Fall time / Drive Strength Control. 00: 2 mA 01: 4 mA 10: 6 mA 11: 8 mA

Register 17. CLK1 Control

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	CLK1_PDN	MS1_INT	MS1_SRC	CLK1_INV	CLK1_SRC[1:0]		CLK1_IDRV[1:0]	
Type	R/W	R/W	R/W	R/W	R/W		R/W	

Reset value = 0000 0000

Bit	Name	Function
7	CLK1_PDN	Clock 1 Power Down. This bit allows powering down the CLK1 output driver to conserve power when the output is unused. 0: CLK1 is powered up. 1: CLK1 is powered down.
6	MS1_INT	MultiSynth 1 Integer Mode. When the MS1 divider is an even integer, this bit can be used to force MS1 into integer mode to improve jitter performance. Not valid for odd integers. Note that the fractional mode is necessary when a delay offset is specified for CLK1. 0: MS1 operates in fractional division mode. 1: MS1 operates in integer mode.
5	MS1_SRC	MultiSynth Source Select for CLK1. 0: Select PLLA as the source for MultiSynth1. 1: Select PLLB (Si5351A/C only) or VCXO (Si5351B only) for MultiSynth1.
4	CLK1_INV	Output Clock 1 Invert. 0: Output Clock 1 is not inverted. 1: Output Clock 1 is inverted.
3:2	CLK1_SRC[1:0]	Output Clock 1 Input Source. These bits determine the input source for CLK1. 00: Select the XTAL as the clock source for CLK1. This option by-passes both synthesis stages (PLL/VCXO & MultiSynth) and connects CLK1 directly to the oscillator which generates an output frequency determined by the XTAL frequency. 01: Select CLKIN as the clock source for CLK1. This by-passes both synthesis stages (PLL/VCXO & MultiSynth) and connects CLK1 directly to the CLKIN input. This essentially creates a buffered output of the CLKIN input. 10: Select Multisynth 0 as the source for CLK1. 11: Select MultiSynth 1 as the source for CLK1. Select this option when using the Si5351 to generate free-running or synchronous clocks.
1:0	CLK1_IDRV[1:0]	CLK1 Output Rise and Fall time / Drive Strength Control. 00: 2 mA 01: 4 mA 10: 6 mA 11: 8 mA

Register 18. CLK2 Control

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	CLK2_PDN	MS2_INT	MS2_SRC	CLK2_INV	CLK2_SRC[1:0]		CLK2_IDRV[1:0]	
Type	R/W	R/W	R/W	R/W	R/W		R/W	

Reset value = 0000 0000

Bit	Name	Function
7	CLK2_PDN	Clock 2 Power Down. This bit allows powering down the CLK2 output driver to conserve power when the output is unused. 0: CLK2 is powered up. 1: CLK2 is powered down.
6	MS2_INT	MultiSynth 2 Integer Mode. When the MS2 divider is an even integer, this bit can be used to force MS2 into integer mode to improve jitter performance. Not valid for odd integers. Note that the fractional mode is necessary when a delay offset is specified for CLK2. 0: MS2 operates in fractional division mode. 1: MS2 operates in integer mode.
5	MS2_SRC	MultiSynth Source Select for CLK2. 0: Select PLLA as the source for MultiSynth2. 1: Select PLLB (Si5351A/C only) or VCXO (Si5351B only) for MultiSynth2.
4	CLK2_INV	Output Clock 2 Invert. 0: Output Clock 2 is not inverted. 1: Output Clock 2 is inverted.
3:2	CLK2_SRC[1:0]	Output Clock 2 Input Source. These bits determine the input source for CLK2. 00: Select the XTAL as the clock source for CLK2. This option by-passes both synthesis stages (PLL/VCXO & MultiSynth) and connects CLK2 directly to the oscillator which generates an output frequency determined by the XTAL frequency. 01: Select CLKIN as the clock source for CLK2. This by-passes both synthesis stages (PLL/VCXO & MultiSynth) and connects CLK2 directly to the CLKIN input. This essentially creates a buffered output of the CLKIN input. 10: Select MultiSynth 0 as the source for CLK2. 11: Select MultiSynth 2 as the source for CLK2. Select this option when using the Si5351 to generate free-running or synchronous clocks.
1:0	CLK2_IDRV[1:0]	CLK2 Output Rise and Fall time / Drive Strength Control. 00: 2 mA 01: 4 mA 10: 6 mA 11: 8 mA

Register 19. CLK3 Control

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	CLK3_PDN	MS3_INT	MS3_SRC	CLK3_INV	CLK3_SRC[1:0]		CLK3_IDRV[1:0]	
Type	R/W	R/W	R/W	R/W	R/W		R/W	

Reset value = 0000 0000

Bit	Name	Function
7	CLK3_PDN	Clock 3 Power Down. This bit allows powering down the CLK3 output driver to conserve power when the output is unused. 0: CLK3 is powered up. 1: CLK3 is powered down. Write this bit to a 1 when programming a Si5351A-B-GT 10-MSOP Device.
6	MS3_INT	MultiSynth 3 Integer Mode. When the MS3 divider is an even integer, this bit can be used to force MS3 into integer mode to improve jitter performance. Not valid for odd integers. Note that the fractional mode is necessary when a delay offset is specified for CLK3. 0: MS3 operates in fractional division mode. 1: MS3 operates in integer mode.
5	MS3_SRC	MultiSynth Source Select for CLK3. 0: Select PLLA as the source for MultiSynth3. 1: Select PLLB (Si5351A/C only) or VCXO (Si5351B only) for MultiSynth3.
4	CLK3_INV	Output Clock 3 Invert. 0: Output Clock 3 is not inverted. 1: Output Clock 3 is inverted.
3:2	CLK3_SRC[1:0]	Output Clock 3 Input Source. These bits determine the input source for CLK3. 00: Select the XTAL as the clock source for CLK3. This option by-passes both synthesis changes (PCL/VCXO and Multisynth) and connects CLK3 directly to the oscillator which generates an output frequency determined by the XTAL frequency. 01: Select CLKIN as the clock source for CLK3. This by-passes both synthesis stages (PLL/VCXO and Multisynth) and connects CLK3 directly to the CLKIN input. This essentially creates a buffered output of the CLKIN input. 10: Select Multisynth 0 as the source for CLK3. 11: Select Multisynth 3 as the source for CLK3. Select this option when using the Si5351 to generate free-running or synchronous clocks.
1:0	CLK3_IDRV[1:0]	CLK3 Output Rise and Fall time / Drive Strength Control. 00: 2 mA 01: 4 mA 10: 6 mA 11: 8 mA

Register 20. CLK4 Control

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	CLK4_PDN	MS4_INT	MS4_SRC	CLK4_INV	CLK4_SRC[1:0]		CLK4_IDRV[1:0]	
Type	R/W	R/W	R/W	R/W	R/W		R/W	

Reset value = 0000 0000

Bit	Name	Function
7	CLK4_PDN	Clock 4 Power Down. This bit allows powering down the CLK4 output driver to conserve power when the output is unused. 0: CLK4 is powered up. 1: CLK4 is powered down. Write this bit to a 1 when programming a Si5351A-B-GT 10-MSOP Device.
6	MS4_INT	MultiSynth 4 Integer Mode. When the MS4 divider is an even integer, this bit can be used to force MS4 into integer mode to improve jitter performance. Not valid for odd integers. Note that the fractional mode is necessary when a delay offset is specified for CLK4. 0: MS4 operates in fractional division mode. 1: MS4 operates in integer mode.
5	MS4_SRC	MultiSynth Source Select for CLK4. 0: Select PLLA as the source for MultiSynth4. 1: Select PLLB (Si5351A/C only) or VCXO (Si5351B only) for MultiSynth4.
4	CLK4_INV	Output Clock 4 Invert. 0: Output Clock 4 is not inverted. 1: Output Clock 4 is inverted.
3:2	CLK4_SRC[1:0]	Output Clock 4 Input Source. These bits determine the input source for CLK4. 00: Select the XTAL as the clock source for CLK4. This option by-passes both synthesis stages (PLL/VCXO & MultiSynth) and connects CLK4 directly to the oscillator which generates an output frequency determined by the XTAL frequency. 01: Select CLKIN as the clock source for CLK4. This by-passes both synthesis stages (PLL/VCXO & MultiSynth) and connects CLK4 directly to the CLKIN input. This essentially creates a buffered output of the CLKIN input. 10: Reserved. Do not select this option. 11: Select MultiSynth 4 as the source for CLK4. Select this option when using the Si5351 to generate free-running or synchronous clocks.
1:0	CLK4_IDRV[1:0]	CLK4 Output Rise and Fall time / Drive Strength Control. 00: 2 mA 01: 4 mA 10: 6 mA 11: 8 mA

Register 21. CLK5 Control

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	CLK5_PDN	MS5_INT	MS5_SRC	CLK5_INV	CLK5_SRC[1:0]		CLK5_IDRV[1:0]	
Type	R/W	R/W	R/W	R/W	R/W		R/W	

Reset value = 0000 0000

Bit	Name	Function
7	CLK5_PDN	Clock 5 Power Down. This bit allows powering down the CLK5 output driver to conserve power when the output is unused. 0: CLK5 is powered up. 1: CLK5 is powered down. Write this bit to a 1 when programming a Si5351A-B-GT 10-MSOP Device.
6	MS5_INT	MultiSynth 5 Integer Mode. When the MS5 divider is an even integer, this bit can be used to force MS5 into integer mode to improve jitter performance. Not valid for odd integers. Note that the fractional mode is necessary when a delay offset is specified for CLK5. 0: MS5 operates in fractional division mode. 1: MS5 operates in integer mode.
5	MS5_SRC	MultiSynth Source Select for CLK5. 0: Select PLLA as the source for MultiSynth5. 1: Select PLLB (Si5351A/C only) or VCXO (Si5351B only) for MultiSynth5.
4	CLK5_INV	Output Clock 5 Invert. 0: Output Clock 5 is not inverted. 1: Output Clock 5 is inverted.
3:2	CLK5_SRC[1:0]	Output Clock 5 Input Source. These bits determine the input source for CLK5. 00: Select the XTAL as the clock source for CLK5. This option by-passes both synthesis stages (PLL/VCXO & MultiSynth) and connects CLK5 directly to the oscillator which generates an output frequency determined by the XTAL frequency. 01: Select CLKIN as the clock source for CLK5. This by-passes both synthesis stages (PLL/VCXO & MultiSynth) and connects CLK5 directly to the CLKIN input. This essentially creates a buffered output of the CLKIN input. 10: Select Multisynth 4 as the source for CLK5. 11: Select MultiSynth 5 as the source for CLK5. Select this option when using the Si5351 to generate free-running or synchronous clocks.
1:0	CLK5_IDRV[1:0]	CLK5 Output Rise and Fall time / Drive Strength Control. 00: 2 mA 01: 4 mA 10: 6 mA 11: 8 mA

Register 22. CLK6 Control

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	CLK6_PDN	FBA_INT	MS6_SRC	CLK6_INV	CLK6_SRC[1:0]		CLK6_IDRV[1:0]	
Type	R/W	R/W	R/W	R/W	R/W		R/W	

Reset value = 0000 0000

Bit	Name	Function
7	CLK6_PDN	Clock 7 Power Down. This bit allows powering down the CLK6 output driver to conserve power when the output is unused. 0: CLK6 is powered up. 1: CLK6 is powered down. Write this bit to a 1 when programming a Si5351A-B-GT 10-MSOP Device.
6	FBA_INT	FBA MultiSynth Integer Mode. This bit can be used to force Feedback A Multisynth into Integer mode to improve jitter performance. Note that the fractional mode is necessary when spread spectrum is specified for any output clocks. 0: MSNA operates in fractional division mode. 1: MSNA operates in integer mode.
5	MS6_SRC	MultiSynth Source Select for CLK6. 0: Select PLLA as the source for MultiSynth6. 1: Select PLLB (Si5351A/C only) or VCXO (Si5351B only) for MultiSynth6.
4	CLK6_INV	Output Clock 6 Invert. 0: Output Clock 6 is not inverted. 1: Output Clock 6 is inverted.
3:2	CLK6_SRC[1:0]	Output Clock 6 Input Source. These bits determine the input source for CLK6. 00: Select the XTAL as the clock source for CLK6. This option by-passes both synthesis stages (PLL/VCXO & MultiSynth) and connects CLK6 directly to the oscillator which generates an output frequency determined by the XTAL frequency. 01: Select CLKIN as the clock source for CLK6. This by-passes both synthesis stages (PLL/VCXO & MultiSynth) and connects CLK6 directly to the CLKIN input. This essentially creates a buffered output of the CLKIN input. 10: Select Multisynth 4 as the source for CLK6. 11: Select MultiSynth 6 as the source for CLK6. Select this option when using the Si5351 to generate free-running or synchronous clocks.
1:0	CLK6_IDRV[1:0]	CLK6 Output Rise and Fall time / Drive Strength Control. 00: 2 mA 01: 4 mA 10: 6 mA 11: 8 mA

Register 23. CLK7 Control

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	CLK7_PDN	FBB_INT	MS7_SRC	CLK7_INV	CLK7_SRC[1:0]		CLK7_IDRV[1:0]	
Type	R/W	R/W	R/W	R/W	R/W		R/W	

Reset value = 0000 0000

Bit	Name	Function
7	CLK7_PDN	Clock 7 Power Down. This bit allows powering down the CLK7 output driver to conserve power when the output is unused. 0: CLK7 is powered up. 1: CLK7 is powered down. Write this bit to a 1 when programming a Si5351A-B-GT 10-MSOP Device.
6	FBB_INT	FBB MultiSynth Integer Mode. This bit can be used to force Feedback B into Integer mode to improve jitter performance. 0: MSNB operates in fractional division mode. 1: MSNB operates in integer mode.
5	MS7_SRC	MultiSynth Source Select for CLK7. 0: Select PLLA as the source for MultiSynth7. 1: Select PLLB (Si5351A/C only) or VCXO (Si5351B only) for MultiSynth7.
4	CLK7_INV	Output Clock 7 Invert. 0: Output Clock 7 is not inverted. 1: Output Clock 7 is inverted.
3:2	CLK7_SRC[1:0]	Output Clock 7 Input Source. These bits determine the input source for CLK7. 00: Select the XTAL as the clock source for CLK7. This option by-passes both synthesis stages (PLL/VCXO & MultiSynth) and connects CLK7 directly to the oscillator which generates an output frequency determined by the XTAL frequency. 01: Select CLKIN as the clock source for CLK7. This by-passes both synthesis stages (PLL/VCXO & MultiSynth) and connects CLK7 directly to the CLKIN input. This essentially creates a buffered output of the CLKIN input. 10: Select Multisynth 4 as the source for CLK7. 11: Select MultiSynth 7 as the source for CLK7. Select this option when using the Si5351 to generate free-running or synchronous clocks.
1:0	CLK7_IDRV[1:0]	CLK7 Output Rise and Fall time / Drive Strength Control. 00: 2 mA 01: 4 mA 10: 6 mA 11: 8 mA

Register 24. CLK3–0 Disable State

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	CLK3_DIS_STATE		CLK2_DIS_STATE		CLK1_DIS_STATE		CLK0_DIS_STATE	
Type	R/W		R/W		R/W		R/W	

Reset value = 0000 0000

Bit	Name	Function
7:0	CLKx_DIS_STATE	Clock x Disable State. Where x = 0, 1, 2, 3. These 2 bits determine the state of the CLKx output when disabled. Individual output clocks can be disabled using register <i>Output Enable Control</i> located at address 3. Outputs are also disabled using the OEB pin. 00: CLKx is set to a LOW state when disabled. 01: CLKx is set to a HIGH state when disabled. 10: CLKx is set to a HIGH IMPEDANCE state when disabled. 11: CLKx is NEVER DISABLED.

Register 25. CLK7–4 Disable State

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	CLK7_DIS_STATE		CLK6_DIS_STATE		CLK5_DIS_STATE		CLK4_DIS_STATE	
Type	R/W		R/W		R/W		R/W	

Reset value = 0000 0000

Bit	Name	Function
7:0	CLKx_DIS_STATE	Clock x Disable State. Where x = 4, 5, 6, 7. These 2 bits determine the state of the CLKx output when disabled. Individual output clocks can be disabled using register <i>Output Enable Control</i> located at address 3. Outputs are also disabled using the OEB pin. 00: CLKx is set to a LOW state when disabled. 01: CLKx is set to a HIGH state when disabled. 10: CLKx is set to a HIGH IMPEDANCE state when disabled. 11: CLKx is NEVER DISABLED.

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Register 26. Multisynth NA Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MSNA_P3[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MSNA_P3[15:8]	Multisynth NA Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the PLLA Feedback Multisynth Divider.

Register 27. Multisynth NA Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MSNA_P3[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MSNA_P3[7:0]	Multisynth NA Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the PLLA Feedback Multisynth Divider.

Register 28. Multisynth NA Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	Unused				Reserved		MSNA_P1[17:16]	
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:4	Unused	Unused.
3:2	Reserved	Reserved. Leave as default, 0.
1:0	MSNA_P1[17:16]	Multisynth NA Parameter 1. This 18-bit number is an encoded representation of the integer part of the PLLA Feedback Multisynth divider.

Register 29. Multisynth NA Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MSNA_P1[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MSNA_P1[15:8]	Multisynth NA Parameter 1. This 18-bit number is an encoded representation of the integer part of the PLLA Feedback Multisynth divider.

Register 30. Multisynth NA Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MSNA_P1[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MSNA_P1[7:0]	Multisynth NA Parameter 1. This 18-bit number is an encoded representation of the integer part of the PLLA Feedback Multisynth divider.

Register 31. Multisynth NA Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MSNA_P3[19:16]				MSNA_P2[19:16]			
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:4	MSNA_P3[19:16]	Multisynth NA Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the PLLA Feedback Multisynth divider.
3:0	MSNA_P2[19:16]	Multisynth NA Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the PLLA Feedback Multisynth divider.

Register 32. Multisynth NA Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MSNA_P2[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MSNA_P2[15:8]	Multisynth NA Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the PLLA Feedback Multisynth divider.

Register 33. Multisynth NA Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MSNA_P2[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MSNA_P2[7:0]	Multisynth NA Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the PLLA Feedback Multisynth divider.

Register 34. Multisynth NB Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MSNB_P3[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MSNB_P3[15:8]	Multisynth NA Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the PLLA Feedback Multisynth divider.

Register 35. Multisynth NB Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MSNB_P3[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MSNB_P3[7:0]	Multisynth NB Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the PLLB Feedback Multisynth divider.

Register 36. Multisynth NB Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	Unused				Reserved		MSNB_P1[17:16]	
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:4	Unused	
3:2	Reserved	Reserved. Leave as default, 0.
1:0	MSNB_P1[17:16]	Multisynth NB Parameter 1. This 18-bit number is an encoded representation of the integer part for the fractional part of the PLLB Feedback Multisynth divider.

Register 37. Multisynth NB Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MSNB_P1[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MSNB_P1[15:8]	Multisynth NB Parameter 1. This 18-bit number is an encoded representation of the integer part of the PLLB Feedback Multisynth divider.

Register 38. Multisynth NB Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MSNB_P1[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MSNB_P1[7:0]	Multisynth NB Parameter 1. This 18-bit number is an encoded representation of the integer part of the PLLB Feedback Multisynth divider.

Register 39. Multisynth NB Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MSNB_P3[19:16]				MSNB_P2[19:16]			
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:4	MSNB_P3[19:16]	Multisynth NB Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the PLLB Feedback Multisynth divider.
3:0	MSNB_P2[19:16]	Multisynth NB Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the PLLB Feedback Multisynth divider.

Register 40. Multisynth NB Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MSNB_P2[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MSNB_P2[15:8]	Multisynth NB Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the PLLB Feedback Multisynth divider.

Register 41. Multisynth NB Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MSNB_P2[7:0]							
Type	R/W							

Bit	Name	Function
7:0	MSNB_P2[7:0]	Multisynth NB Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the PLLB Feedback Multisynth divider.

Register 42. Multisynth0 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS0_P3[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS0_P3[15:8]	Multisynth0 Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the MultiSynth0 Divider.

Register 43. Multisynth0 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS0_P3[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS0_P3[7:0]	Multisynth0 Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the MultiSynth0 Divider.

Register 44. Multisynth0 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name		R0_DIV[2:0]			MS0_DIVBY4[1:0]		MS0_P1[17:16]	
Type	R/W		R/W		R/W	R/W		R/W

Reset value = xxxx xxxx

Bit	Name	Function
7	Unused	
6:4	R0_DIV[2:0]	R0 Output Divider. 000b: Divide by 1 001b: Divide by 2 010b: Divide by 4 011b: Divide by 8 100b: Divide by 16 101b: Divide by 32 110b: Divide by 64 111b: Divide by 128
3:2	MS0_DIVBY4[1:0]	MS0 Divide by 4 Enable. 11: Divide by 4 enabled. 00: Divide by a value other than 4.
1:0	MS0_P1[17:16]	Multisynth0 Parameter 1. This 18-bit number is an encoded representation of the integer part of the Multi-Synth0 divider.

Register 45. Multisynth0 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS0_P1[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS0_P1[15:8]	Multisynth0 Parameter 1. This 18-bit number is an encoded representation of the integer part of the Multi-Synth1 divider.

Register 46. Multisynth0 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS0_P1[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS0_P1[7:0]	Multisynth0 Parameter 1. This 18-bit number is an encoded representation of the integer part of the MultiSynth1 divider.

Register 47. Multisynth0 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS0_P3[19:16]				MS0_P2[19:16]			
Type	R/W				R/W			

Reset value = xxxx xxxx

Bit	Name	Function
7:4	MS0_P3[19:16]	Multisynth0 Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the MultiSynth0 Divider
3:0	MS0_P2[19:16]	Multisynth0 Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the MultiSynth1 Divider.

Register 48. Multisynth0 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS0_P2[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS0_P2[15:8]	Multisynth0 Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the MultiSynth1 Divider.

Register 49. Multisynth0 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS0_P2[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS0_P2[7:0]	Multisynth0 Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the MultiSynth1 Divider.

Register 50. Multisynth1 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS1_P3[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS1_P3[15:8]	Multisynth1 Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the MultiSynth1 Divider.

Register 51. Multisynth1 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS1_P3[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS1_P3[7:0]	Multisynth1 Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the MultiSynth1 Divider.

Register 52. Multisynth1 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name		R1_DIV[2:0]			MS1_DIVBY4[1:0]		MS1_P1[17:16]	
Type	R/W		R/W		R/W	R/W		R/W

Reset value = xxxx xxxx

Bit	Name	Function
7	Unused	
6:4	R1_DIV[2:0]	R1 Output Divider. 000b: Divide by 1 001b: Divide by 2 010b: Divide by 4 011b: Divide by 8 100b: Divide by 16 101b: Divide by 32 110b: Divide by 64 111b: Divide by 128
3:2	MS1_DIVBY4[1:0]	MS1 Divide by 4 Enable. 11: Divide by 4 enabled. 00: Divide by a value other than 4.
1:0	MS1_P1[17:16]	Multisynth1 Parameter 1. This 18-bit number is an encoded representation of the integer part of the Multi-Synth1 divider.

Register 53. Multisynth1 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS1_P1[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS1_P1[15:8]	Multisynth1 Parameter 1. This 18-bit number is an encoded representation of the integer part of the Multi-Synth1 divider.

Register 54. Multisynth1 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS1_P1[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS1_P1[7:0]	Multisynth1 Parameter 1. This 18-bit number is an encoded representation of the integer part of the Multi-Synth1 divider.

Register 55. Multisynth1 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS1_P3[19:16]				MS1_P2[19:16]			
Type	R/W				R/W			

Reset value = xxxx xxxx

Bit	Name	Function
7:4	MS1_P3[19:16]	Multisynth1 Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the Multisynth1 Divider
3:0	MS1_P2[19:16]	Multisynth1 Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the MultiSynth1 Divider.

Register 56. Multisynth1 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS1_P2[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS1_P2[15:8]	Multisynth1 Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the MultiSynth1 divider.

Register 57. Multisynth1 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS1_P2[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS1_P2[7:0]	Multisynth1 Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the MultiSynth1 divider.

Register 58. Multisynth2 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS2_P3[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS2_P3[15:8]	Multisynth2 Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the MultiSynth2 divider.

Register 59. Multisynth2 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS2_P3[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS2_P3[7:0]	Multisynth2 Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the MultiSynth2 divider.

Register 60. Multisynth2 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name		R2_DIV[2:0]			MS2_DIVBY4[1:0]		MS2_P1[17:16]	
Type	R/W	R/W			R/W	R/W	R/W	

Reset value = xxxx xxxx

Bit	Name	Function
7	Unused	
6:4	R2_DIV[2:0]	R2 Output Divider. 000b: Divide by 1 001b: Divide by 2 010b: Divide by 4 011b: Divide by 8 100b: Divide by 16 101b: Divide by 32 110b: Divide by 64 111b: Divide by 128
3:2	MS2_DIVBY4[1:0]	MS2 Divide by 4 Enable. 11: Divide by 4 enabled. 00: Divide by a value other than 4.
1:0	MS2_P1[17:16]	Multisynth2 Parameter 1. This 18-bit number is an encoded representation of the integer part of the Multisynth2 divider.

Register 61. Multisynth2 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS2_P1[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS2_P1[15:8]	Multisynth2 Parameter 1. This 18-bit number is an encoded representation of the integer part of the Multisynth2 divider.

Register 62. Multisynth2 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS2_P1[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS2_P1[7:0]	Multisynth2 Parameter 1. This 18-bit number is an encoded representation of the integer part of the Multisynth2 divider.

Register 63. Multisynth2 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS2_P3[19:16]				MS2_P2[19:16]			
Type	R/W				R/W			

Reset value = xxxx xxxx

Bit	Name	Function
7:4	MS2_P3[19:16]	Multisynth2 Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the Multisynth2 divider
3:0	MS2_P2[19:16]	Multisynth2 Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the Multisynth2 divider.

Register 64. Multisynth2 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS2_P2[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS2_P2[15:8]	Multisynth2 Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the Multisynth2 divider.

Register 65. Multisynth2 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS2_P2[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS2_P2[7:0]	Multisynth2 Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the Multisynth2 divider.

Register 66. Multisynth3 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS3_P3[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS3_P3[15:8]	Multisynth3 Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the Multisynth3 divider.

Register 67. Multisynth3 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS3_P3[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS3_P3[7:0]	Multisynth3 Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the Multisynth3 divider.

Register 68. Multisynth3 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name		R3_DIV[2:0]			MS3_DIVBY4[1:0]		MS3_P1[17:16]	
Type	R/W	R/W			R/W	R/W	R/W	

Reset value = xxxx xxxx

Bit	Name	Function
7	Unused	
6:4	R3_DIV[2:0]	R3 Output Divider. 000b: Divide by 1 001b: Divide by 2 010b: Divide by 4 011b: Divide by 8 100b: Divide by 16 101b: Divide by 32 110b: Divide by 64 111b: Divide by 128
3:2	MS3_DIVBY4[1:0]	MS3 Divide by 4 Enable. 11: Divide by 4 enabled. 00: Divide by a value other than 4.
1:0	MS3_P1[17:16]	Multisynth3 Parameter 1. This 18-bit number is an encoded representation of the integer part of the Multisynth3 divider.

Register 69. Multisynth3 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS3_P1[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS3_P1[15:8]	Multisynth3 Parameter 1. This 18-bit number is an encoded representation of the integer part of the Multisynth3 divider.

Register 70. Multisynth3 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS3_P1[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS3_P1[7:0]	Multisynth3 Parameter 1. This 18-bit number is an encoded representation of the integer part of the Multisynth3 divider.

Register 71. Multisynth3 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS3_P3[19:16]				MS3_P2[19:16]			
Type	R/W				R/W			

Reset value = xxxx xxxx

Bit	Name	Function
7:4	MS3_P3[19:16]	Multisynth3 Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the Multisynth3 divider
3:0	MS3_P2[19:16]	Multisynth3 Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the Multisynth3 divider.

Register 72. Multisynth3 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS3_P2[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS3_P2[15:8]	Multisynth3 Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the Multisynth3 divider.

Register 73. Multisynth3 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS3_P2[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS3_P2[7:0]	Multisynth3 Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the Multisynth3 divider.

Register 74. Multisynth4 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS4_P3[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS4_P3[15:8]	Multisynth4 Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the Multisynth4 divider.

Register 75. Multisynth4 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS4_P3[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS4_P3[7:0]	Multisynth4 Parameter 3. This 20-bit number is an encoded representation of the numerator for the fractional part of the Multisynth4 divider.

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Register 76. Multisynth4 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name		R4_DIV[2:0]			MS4_DIVBY4[1:0]		MS4_P1[17:16]	
Type	R/W	R/W			R/W	R/W	R/W	

Reset value = xxxx xxxx

Bit	Name	Function
7	Unused	
6:4	R4_DIV[2:0]	R4 Output Divider. 000b: Divide by 1 001b: Divide by 2 010b: Divide by 4 011b: Divide by 8 100b: Divide by 16 101b: Divide by 32 110b: Divide by 64 111b: Divide by 128
3:2	MS4_DIVBY4[1:0]	MS4 Divide by 4 Enable. 11: Divide by 4 enabled. 00: Divide by a value other than 4.
1:0	MS4_P1[17:16]	Multisynth4 Parameter 1. This 18-bit number is an encoded representation of the integer part of the Multisynth4 divider.

Register 77. Multisynth4 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS4_P1[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS4_P1[15:8]	Multisynth4 Parameter 1. This 18-bit number is an encoded representation of the integer part of the Multisynth4 divider.

Register 78. Multisynth4 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS4_P1[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS4_P1[7:0]	Multisynth4 Parameter 1. This 18-bit number is an encoded representation of the integer part of the Multisynth4 divider.

Register 79. Multisynth4 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS4_P3[19:16]				MS4_P2[19:16]			
Type	R/W				R/W			

Reset value = xxxx xxxx

Bit	Name	Function
7:4	MS4_P3[19:16]	Multisynth4 Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the Multisynth4 divider
3:0	MS4_P2[19:16]	Multisynth4 Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the Multisynth4 divider.

Register 80. Multisynth4 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS4_P2[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS4_P2[15:8]	Multisynth4 Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the Multisynth4 Divider.

Register 81. Multisynth4 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS4_P2[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS4_P2[7:0]	Multisynth4 Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the Multisynth4 divider.

Register 82. Multisynth5 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS5_P3[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS5_P3[15:8]	Multisynth5 Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the Multisynth5 divider.

Register 83. Multisynth5 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS5_P3[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS5_P3[7:0]	Multisynth5 Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the Multisynth5 divider.

Register 84. Multisynth5 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name		R5_DIV[2:0]			MS5_DIVBY4[1:0]		MS5_P1[17:16]	
Type	R/W	R/W			R/W	R/W	R/W	

Reset value = xxxx xxxx

Bit	Name	Function
7	Unused	
6:4	R5_DIV[2:0]	R5 Output Divider. 000b: Divide by 1 001b: Divide by 2 010b: Divide by 4 011b: Divide by 8 100b: Divide by 16 101b: Divide by 32 110b: Divide by 64 111b: Divide by 128
3:2	MS5_DIVBY4[1:0]	MS5 Divide by 4 Enable. 11: Divide by 4 enabled. 00: Divide by a value other than 4.
1:0	MS5_P1[17:16]	Multisynth5 Parameter 1. This 18-bit number is an encoded representation of the integer part of the Multisynth5 divider.

Register 85. Multisynth5 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS5_P1[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS5_P1[15:8]	Multisynth5 Parameter 1. This 18-bit number is an encoded representation of the integer part of the Multisynth5 divider.

Register 86. Multisynth5 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS5_P1[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS5_P1[7:0]	Multisynth5 Parameter 1. This 18-bit number is an encoded representation of the integer part of the Multisynth5 divider.

Register 87. Multisynth5 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS5_P3[19:16]				MS5_P2[19:16]			
Type	R/W				R/W			

Reset value = xxxx xxxx

Bit	Name	Function
7:4	MS5_P3[19:16]	Multisynth5 Parameter 3. This 20-bit number is an encoded representation of the denominator for the fractional part of the Multisynth5 divider
3:0	MS5_P2[19:16]	Multisynth5 Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the MultiSynth5 divider.

Register 88. Multisynth5 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS5_P2[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS5_P2[15:8]	Multisynth5 Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the Multisynth5 Divider.

Register 89. Multisynth5 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS5_P2[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS5_P2[7:0]	Multisynth5 Parameter 2. This 20-bit number is an encoded representation of the numerator for the fractional part of the Multisynth5 Divider.

Register 90. Multisynth6 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS6_P1[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS6_P1[7:0]	Multisynth6 Parameter 1. This 8-bit number is the Multisynth6 divide ratio. Multisynth6 divide ratio can only be even integers greater than or equal to 6. All other divide values are invalid.

Register 91. Multisynth7 Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	MS7_P1[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	MS7_P1[7:0]	Multisynth7 Parameter 1. This 8-bit number is the Multisynth6 divide ratio. Multisynth6 divide ratio can only be even integers greater than or equal to 6. All other divide values are invalid.

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Register 92. Clock 6 and 7 Output Divider

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	Reserved	R7_DIV[2:0]			Reserved	R6_DIV[2:0]		
Type	R/W	R/W			R/W	R/W		

Reset value = xxxx xxxx

Bit	Name	Function
7	Reserved	Leave as default.
6:4	R7_DIV[2:0]	R7 Output Divider. 000b: Divide by 1 001b: Divide by 2 010b: Divide by 4 011b: Divide by 8 100b: Divide by 16 101b: Divide by 32 110b: Divide by 64 111b: Divide by 128
3	Reserved	Leave as default.
1:0	R6_DIV[2:0]	R6 Output Divider. 000b: Divide by 1 001b: Divide by 2 010b: Divide by 4 011b: Divide by 8 100b: Divide by 16 101b: Divide by 32 110b: Divide by 64 111b: Divide by 128

Register 149. Spread Spectrum Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	SSC_EN	SSDN_P2[14:8]						
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7	SSC_EN	Spread Spectrum Enable 1: Enable 0: Disable Note: Spread Spectrum Enable control pin is available on the Si5351A and B devices. Spread spectrum enable functionality is a logical OR of the SSEN pin and SSC_EN register bit, so for the SSEN pin to work properly, this register bit must be set to 0.
6:0	SSDN_P2[14:8]	PLL A Spread Spectrum Down Parameter 2.

Register 150. Spread Spectrum Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	SSDN_P2[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	SSDN_P2[7:0]	PLL A Spread Spectrum Down Parameter 2.

Register 151. Spread Spectrum Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	SSC_MODE	SSDN_P3[14:8]						
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7	SSC_MODE	Spread Spectrum Mode. 0: Down Spread 1: Center Spread
6:0	SSDN_P3[14:8]	PLL A Spread Spectrum Down Parameter 3.

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Register 152. Spread Spectrum Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	SSDN_P3[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	SSDN_P3[7:0]	PLL A Spread Spectrum Down Parameter 3.

Register 153. Spread Spectrum Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	SSDN_P1[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	SSDN_P1[7:0]	PLL A Spread Spectrum Down Parameter 1.

Register 154. Spread Spectrum Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	SSUDP[11:8]				SSDN_P1[11:8]			
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:4	SSUDP[11:8]	PLL A Spread Spectrum Up/Down Parameter.
3:0	SSDN_P1[11:8]	PLL A Spread Spectrum Down Parameter 1.

Register 155. Spread Spectrum Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	SSUDP[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	SSUDP[7:0]	PLL A Spread Spectrum Up/Down Parameter.

Register 156. Spread Spectrum Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	Unused	SSUP_P2[14:8]						
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7	Unused	Unused.
6:0	SSUP_P2[14:8]	PLL A Spread Spectrum Up Parameter 2.

Register 157. Spread Spectrum Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	SSUP_P2[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	SSUP_P2[7:0]	PLL A Spread Spectrum Up Parameter 2.

Register 158. Spread Spectrum Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	Unused	SSUP_P3[14:8]						
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7	Unused	Unused.
6:0	SSUP_P3[14:8]	PLL A Spread Spectrum Up Parameter 3.

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Register 159. Spread Spectrum Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	SSUP_P3[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	SSUP_P3[7:0]	PLL A Spread Spectrum Up Parameter 3.

Register 160. Spread Spectrum Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	SSUP_P1[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	SSUP_P1[7:0]	PLL A Spread Spectrum Up Parameter 1.

Register 161. Spread Spectrum Parameters

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	SS_NCLK[3:0]				SSUP_P1[11:8]			
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:4	SS_NCLK[3:0]	Must write 0000b to these bits.
3:0	SSUP_P1[11:8]	PLL A Spread Spectrum Up Parameter 1.

Register 162. VCXO Parameter

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	VCXO_Param[7:0]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	VCXO_Param[7:0]	VCXO Parameter.

Register 163. VCXO Parameter

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	VCXO_Param[15:8]							
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:0	VCXO_Param[15:8]	VCXO Parameter.

Register 164. VCXO Parameter

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	Reserved		VCXO_Param[21:16]					
Type	R/W							

Reset value = xxxx xxxx

Bit	Name	Function
7:6	Reserved	Reserved. Only write 00b to these bits.
5:0	VCXO_Param[21:16]	VCXO Parameter.

Register 165. CLK0 Initial Phase Offset

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	CLK0_PHOFF[6:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Reset value = 0000 0000

Bit	Name	Function
7	Reserved	Only write 0 to this bit.
6:0	CLK0_PHOFF[6:0]	Clock 0 Initial Phase Offset. CLK0_PHOFF[6:0] is an unsigned integer with one LSB equivalent to a time delay of $T_{vco}/4$, where T_{vco} is the period of the VCO/PLL associated with this output.

Register 166. CLK1 Initial Phase Offset

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	CLK1_PHOFF[6:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Reset value = 0000 0000

Bit	Name	Function
7	Reserved	Only write 0 to this bit.
6:0	CLK1_PHOFF[6:0]	Clock 1 Initial Phase Offset. CLK1_PHOFF[6:0] is an unsigned integer with one LSB equivalent to a time delay of $T_{vco}/4$, where T_{vco} is the period of the VCO/PLL associated with this output.

Register 167. CLK2 Initial Phase Offset

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	CLK2_PHOFF[6:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Reset value = 0000 0000

Bit	Name	Function
7	Reserved	Only write 0 to this bit.
6:0	CLK2_PHOFF[6:0]	Clock 2 Initial Phase Offset. CLK2_PHOFF[6:0] is an unsigned integer with one LSB equivalent to a time delay of $T_{vco}/4$, where T_{vco} is the period of the VCO/PLL associated with this output.

Register 168. CLK3 Initial Phase Offset

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	CLK3_PHOFF[6:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Reset value = 0000 0000

Bit	Name	Function
7	Reserved	Only write 0 to this bit.
6:0	CLK3_PHOFF[6:0]	Clock 3 Initial Phase Offset. CLK3_PHOFF[6:0] is an unsigned integer with one LSB equivalent to a time delay of $T_{vco}/4$, where T_{vco} is the period of the VCO/PLL associated with this output.

Register 169. CLK4 Initial Phase Offset

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	CLK4_PHOFF[6:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Reset value = 0000 0000

Bit	Name	Function
7	Reserved	Only write 0 to this bit.
6:0	CLK4_PHOFF[6:0]	Clock 4 Initial Phase Offset. CLK4_PHOFF[6:0] is an unsigned integer with one LSB equivalent to a time delay of $T_{vco}/4$, where T_{vco} is the period of the VCO/PLL associated with this output.

Register 170. CLK5 Initial Phase Offset

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	CLK5_PHOFF[6:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Reset value = 0000 0000

Bit	Name	Function
7	Reserved	Only write 0 to this bit.
6:0	CLK5_PHOFF[6:0]	Clock 5 Initial Phase Offset. CLK5_PHOFF[6:0] is an unsigned integer with one LSB equivalent to a time delay of $T_{vco}/4$, where T_{vco} is the period of the VCO/PLL associated with this output.

Register 177. PLL Reset

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	PLLB_RST		PLLA_RST					
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Reset value = 0000 0000

Bit	Name	Function
7	PLLB_RST	PLLB_Reset. Writing a 1 to this bit will reset PLLB. This is a self clearing bit.
6	Reserved	Leave as default.
5	PLLA_RST	PLLA_Reset. Writing a 1 to this bit will reset PLLA. This is a self clearing bit.
4:0	Reserved	Leave as default.

Register 183. Crystal Internal Load Capacitance

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	XTAL_CL[1:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Reset value = 11xx xxxx

Bit	Name	Function
7:6	XTAL_CL[1:0]	Crystal Load Capacitance Selection. These 2 bits determine the internal load capacitance value for the crystal. See the Crystal Inputs section in the Si5351 data sheet. 00: Reserved. Do not select this option. 01: Internal CL = 6 pF. 10: Internal CL = 8 pF. 11: Internal CL = 10 pF (default).
5:0	Reserved	Bits 5:0 should be written to 010010b.

Register 187. Fanout Enable

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Name	CLKIN_FA-NOUT_EN	XO_FA-NOUT_EN	Reserved	MS_FA-NOUT_EN	Reserved			
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Reset value = xxxx xxxx

Bit	Name	Function
7	CLKIN_FA-NOUT_EN	Enable fanout of CLKIN to clock output multiplexers. Set this bit to 1b.
6	XO_FA-NOUT_EN	Enable fanout of XO to clock output multiplexers. Set this bit to 1b.
5	Reserved	Reserved.
4	MS_FA-NOUT_EN	Enable fanout of Multisynth0 and Multisynth4 to all output multiplexers. Set this bit to 1b.

DOCUMENT CHANGE LIST

Revision 0.1 to Revision 0.2

- Corrected Typos
 - Feedback fractional divide value can be up to 90 (not 900).
 - Corrected miscellaneous typographical errors.
- Added note to “3.2. Feedback Multisynth Divider Equations” to set MSNB a + b/c divide ratio such that $c = 106$ when using VCXO equations in this document.
- Added Figure 2.

Revision 0.2 to Revision 0.3

- Updated first bullet in “Section 5. Configuring Spread Spectrum Register Parameters.”
- Updated “Section 8.1 Register Map Summary.”
- Updated Register 183.

Revision 0.3 to Revision 0.4

- Corrected typographic error in maximum spread spectrum percentage for center spread on page 8.
- Added rounding operator for up spread depth calculation on page 9.
- Corrected register description errata on pages 11, 13, 31, and 32.

Revision 0.4 to Revision 0.5

- Updated “8.1. Register Map Summary”.
- Updated “6. Configuring Initial Phase Offset Register Parameters”.
- Corrected Register 16 description.

Revision 0.5 to Revision 0.6

- Updated “8.1. Register Map Summary”.
- Corrected Register 161 description.

Revision 0.6 to Revision 0.7

- Corrected Typo in Section “3.1.2. CMOS Clock Source”
- Corrected Typo in Section “3.2. Feedback Multisynth Divider Equations” Now $c = 10^6$
- Removed Section 3.3. Miscellaneous PLL Parameters.
- Fixed minor typos in Section “5. Configuring Spread Spectrum Register Parameters”
- Corrected Register 36 fields in Section “8.1. Register Map Summary”.
- Updated Register 0 description to include the Revision ID.
- Fixed Register 32 Register Name

- Updated Register 36 description

Revision 0.7 to Revision 0.8

- Updated to include ClockBuilder Pro
- Clarified 10-MSOP and 20-QFN package coverage
- Added reference to AN1234 for 16-QFN Programming Instructions
- Updated Section “8.1. Register Map Summary” on page 10.
- Cleaned up register descriptions



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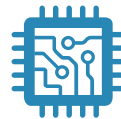
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